

CSE 6242: Team 102 Project Progress Report

1 Introduction

Efficient emergency response is vital to public safety, yet modern 911 systems face rising call volumes, uneven demand, and complex dispatch logistics. In cities like Seattle, even small inefficiencies in triage or unit deployment can cause significant delays. This project leverages Seattle's publicly available **911 data** to model and visualize these dynamics through a data-driven framework integrating machine learning and visual analytics. Using an XGBoost model for response time prediction and Isolation Forest detection method for identifying irregular dispatch patterns, we build an interactive dashboard that delivers interpretable insights to enhance response efficiency, equity, and situational awareness in urban emergency management.

2 Problem Definition

Our objective is to build a smart, visual system that analyzes historical call data to tell city planners and dispatchers exactly where and when to position police and CARE units to ensure the fastest, fairest, and most effective emergency response for every neighborhood. Given a high-volume, temporally dynamic, and spatially variant dataset of 911 calls (C) and corresponding dispatch records (D) in Seattle, the project seeks to achieve the following:

Prediction (Demand Forecasting): Develop an XGBoost model to forecast **Target Response Time** (T_R) for incoming calls based on location (L), priority (P), and time (t). The primary objective is to build a model that minimizes the root mean square error (RSME) between the actual and predicted response times across all calls.

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n \left(T_{R,i}^{\text{actual}} - T_{R,i}^{\text{predicted}} \right)^2}$$

Anomaly Detection (System Monitoring): Implement an unsupervised learning method (Isolation Forest) on the feature space of dispatch events to identify **Abnormal Dispatch Patterns** (Anomaly = 1) and Systemic Inefficiencies (Inefficiency = 1). The objective is to identify rare, high-leverage data points that deviate from the normal operating distribution.

Identify $\{c \in C \mid \text{Isolation Score}(c) > \text{Threshold}\}$

Actionable Synthesis: Design and implement a unified, interactive visual analytics dashboard that translates complex model outputs into actionable insights for operational process improvement and proactive demand management. This synthesis involves displaying the forecasting model's output using Temporal-Spatial Heat Animations to show predicted call surges, and integrating the anomaly detection results via an Incident Flow Timeline to highlight systemic bottlenecks for investigation.

3 Literature Review

Recent research emphasizes the growing role of spatiotemporal modeling and visual analytics in improving emergency response and public safety management. Previous studies such as *Space-Time Urban Explorer* [1] and *Optimizing Local Police Practices* [2] demonstrate how interactive dashboards and visual exploration tools can enhance operational insights for law enforcement, yet they lack predictive integration. Building on this foundation, *Spatiotemporal Predictive Geo-Visualization for Real-Time Systems* [3] and *Ada-GCNLSTM* [4] apply deep learning and graph-based methods to forecast spatial crime distributions, achieving high accuracy but at significant computational cost.

Traditional forecasting approaches, including ARIMA, Holt-Winters, and regression models, have also been explored for EMS and 911 call centers [5, 6], showcasing reasonable short-term performance but limited spatial generalization. Recent healthcare analytics research extends these ideas using scalable machine learning pipelines such as Random Forests for near real-time performance prediction [7] and explainable multi-model regression frameworks for emergency forecasting [8], underscoring the importance of model interpretability and continuous retraining. Optimization and clustering studies [9, 10] further highlight data-driven strategies for officer deployment and response time, while online decision-theoretic frameworks [11] and next-generation computer-aided dispatch systems [12] demonstrate how automation can improve dispatch efficiency. However, administrative and political factors often complicate adoption, as shown in early analyses of CAD implementation [13]. Other works on simulation and algorithm governance [14, 15] emphasize the importance of interpretability and accountability in operational decision support.

These studies highlight the potential of data-driven strategies to improve emergency response and resource allocation. Our project builds on this foundation by integrating lightweight machine-learning models with interactive visualizations to analyze 911 response patterns, synthesizing predictive analytics and decision-support dashboards for practical public safety insights.

4 Proposed Method

Our project integrates two analytical models with aligned visualizations to generate actionable insights for Seattle’s 911 operations (Table 1 and Table 2).

4.1 Data Preprocessing

The dataset underwent several preprocessing steps to ensure quality and relevance. Initial exploration led to the discovery of improperly formatted records causing ingestion errors which were subsequently removed. Further, duplicate entries were eliminated to avoid redundancy. Records with similar descriptive attributes were aggregated in alignment with analytical goals. Outliers were systematically identified and excluded in accordance with an interquartile range (IQR) methodology to maintain data consistency. “Co” and “Care” program metrics were unavailable prior to late October 2023, necessitating the removal of prior records to preserve dataset completeness. Additionally, community response-related data were excluded due to scope limitations, while niche and low-volume categorical records were removed to focus on high impact categories that would be more representative.

4.2 Model 1

It begins with a cleaned dataset of emergency call records containing details such as call type, location, time of day, number of officers dispatched, and priority level. The goal is to estimate each call’s total service time — the duration from dispatch to completion — using a machine learning model. To achieve this, we employed XGBoost, a gradient boosting algorithm known for its strong performance on structured data. After training and evaluating the model using standard regression metrics such as RMSE.

The visualization (Figure 1) highlights the features that most strongly influence the model’s predictions of 911 response times. The number of officers dispatched stands out as the most dominant factor, suggesting that larger response teams are closely associated with longer or more complex service times. The pseudo priority score also plays a major role, reflecting how the urgency or severity of a call impacts the overall duration of response efforts. Several dispatch neighborhoods and sectors—such as Downtown Commercial, King, and Edward—appear among the top features, indicating that location-based characteristics significantly affect response dynamics, likely due to traffic, population density, or incident frequency. Meanwhile, variables like call type (e.g., ONVIEW or ALARM CALL) and priority level contribute moderately, hinting that both

operational classification and situational urgency influence response patterns.

4.3 Model 2

Following predictive modeling, this section introduces a comprehensive anomaly detection framework to diagnose operational irregularities in the 911 data, complementing the forecasting model. Unlike prediction, which estimates expected outcomes, this approach identifies unexpected or atypical behaviors across key dispatch operations, including unusual response times, abrupt call volume changes, and potential routing errors. The methodology performs contextualized anomaly detection to overcome the limitations of generic statistical models. Since normal call volume in one district may be an anomaly elsewhere, the data is segmented into relevant operational groups (call type or sector). This ensures that deviations are measured against comparable baselines, reducing false positives and dramatically improving interpretability in systems like emergency response networks. The framework comprises multiple analytical modules including Temporal Analysis (identifies sudden "bursts" or fluctuations in call volume), Response Time Analysis (uses z-scores to flag atypically long or short service durations relative to peers), and Spatial Analysis (flags calls outside expected geographic clusters (geocoding errors) and checks the alignment between incident location and assigned dispatch sector (routing errors)). Each stream generates an anomaly indicator, aggregated into a composite index for system monitoring. The outputs are visualized using Incident Flow Timelines and spatial heatmaps, providing interpretable feedback to inform data quality checks, policy evaluation, and resource management strategies.

Task	Best Model	Outcome
Response Time Prediction	XGBoost	Reduce average dispatch response time through predictive modeling of call priority and location.
Anomaly Detection	Isolation Forest	Identify systemic inefficiencies or abnormal dispatch patterns for process improvement.

Table 1: Summary of analytical models and expected outcomes for the Seattle 911 Triage/Planner project.

Visualization	Description	Insight Provided
Temporal-Spatial Heat Animation (Figure 2 and Figure 3)	Animated heatmap of call frequency and response time by hour or day.	Reveals city-wide demand cycles and temporal hotspots.
Incident Flow Timeline (Figure 4)	Time-series of 911 calls color-coded by incident type.	Highlights patterns and surges across different categories and time periods.

Table 2: Proposed interactive visualizations for the Seattle 911 Triage/Planner dashboard.

5 Evaluation

5.1 Model 1

For Model 1, the response time prediction framework uses the XGBoost algorithm to estimate the response duration based on incident and dispatch features. Because this is a supervised learning problem with observable ground-truth response times, the evaluation focuses on quantitative performance and operational interpretability. Specifically, root mean squared error (RMSE) is used to measure accuracy across training and test splits, ensuring generalizability and robustness. Beyond

numerical metrics, interpretability is assessed through analysis of feature importances to determine whether key predictors—such as call type, priority, and dispatch sector—correspond with domain knowledge of emergency response behavior. Visualization of residuals over time and geography further helps diagnose systematic bias or underperforming regions. This combination of statistical and operational evaluation provides both empirical validation and practical understanding of how response time patterns can inform future resource allocation and performance optimization within Seattle’s 911 system.

5.2 Model 2

For Model 2, the anomaly detection framework, evaluation presents additional challenges because it is designed as an unsupervised learning approach rather than a predictive model. Because the dataset does not contain predefined or labeled anomalies for direct validation, the evaluation will instead focus on assessing the validity, coherence, and interpretability of the detected anomalies rather than traditional accuracy metrics. Over the next several weeks, prior to the final project submission, the evaluation process will incorporate a combination of internal consistency metrics and qualitative analyses. Specifically, the team will examine the proportion of anomalies detected across modules, the temporal and spatial clustering of anomalies in relation to known operational events, and the degree of overlap between different anomaly types (e.g., response time and routing anomalies). Visual inspection through Incident Flow Timelines and spatial heatmaps will also be used to confirm whether the identified anomalies align with realistic patterns of emergency activity. This iterative evaluation strategy will help establish the robustness and operational relevance of the model in the absence of labeled ground truth data.

6 Conclusions and Discussion

In summary, this project integrates predictive analytics and anomaly detection to enhance operational efficiency in Seattle’s 911 system. The XGBoost model forecasts response times based on call priority and location, while the Isolation Forest framework identifies systemic inefficiencies and abnormal dispatch patterns. These analytical outputs are visualized through interactive temporal–spatial heatmaps and incident flow timelines, transforming data into actionable insights. Ongoing evaluation focuses on interpretability, consistency, and real-world alignment, guiding future refinement toward a scalable decision-support dashboard for emergency resource management [2].

Project Planning

All members will jointly handle pipeline development, code reviews, documentation, and visual design to ensure cross-functional learning and consistency (Table 3).

Week — Focus	All-team activities (all five members contribute each week)
W5 (Nov 2–8)	Evaluate model accuracy; finalize quantitative analyses.
W6 (Nov 9–15)	Merge both models visuals to create and then refine dashboard UX. Begin poster and report drafting.
W7 (Nov 16–21)	Final integration and submission: Freeze codebase, complete poster, README, and demo video; rehearse presentation; submit final deliverables.

Table 3: Three-week project timeline (Nov 2–Nov 21) from progress report to final report.

All team members have contributed an equal amount of effort.

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A Figures

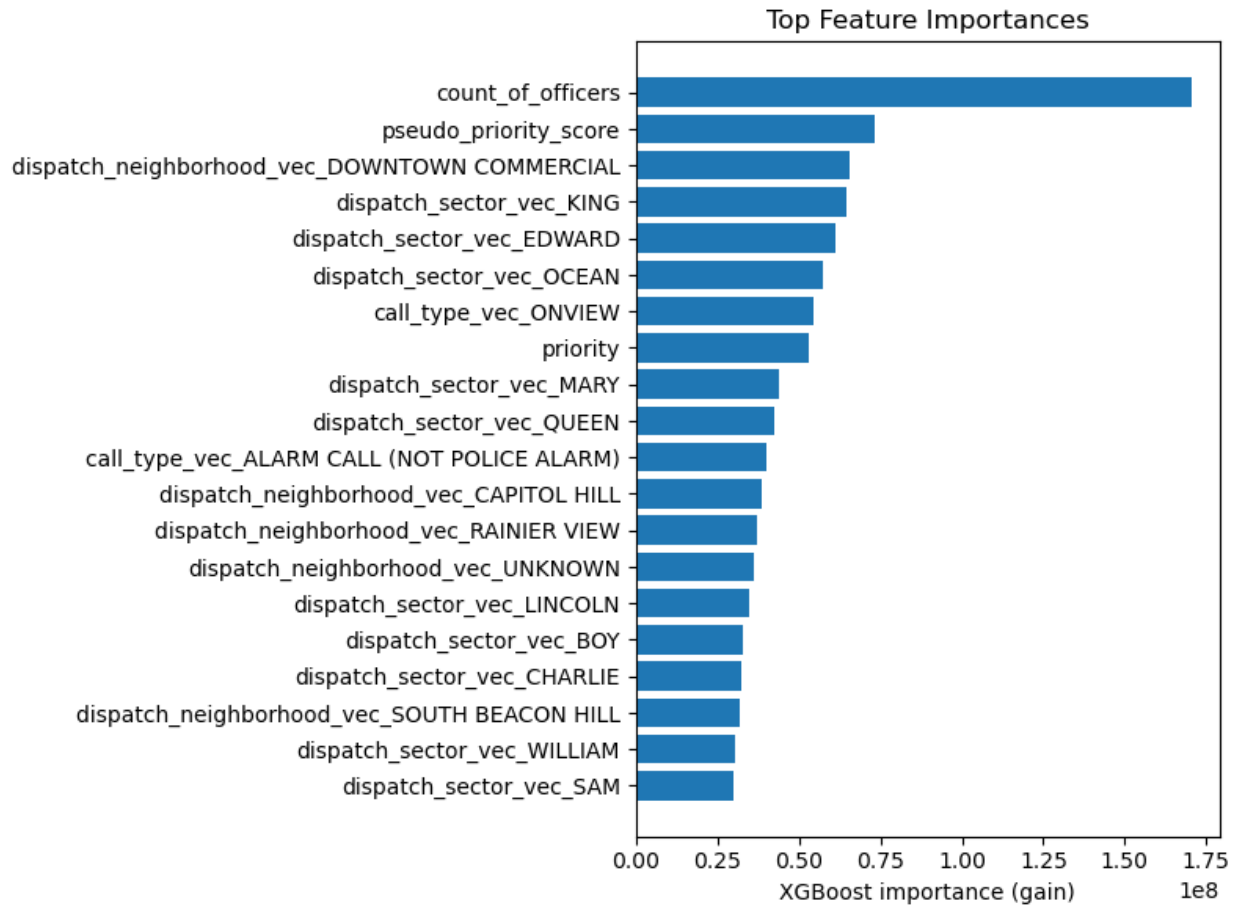


Figure 1: Feature importance ranking showing the most influential variables in predicting SPD response times. Factors such as the number of officers dispatched, pseudo-priority score, and neighborhood or sector identifiers contribute most significantly to model performance.

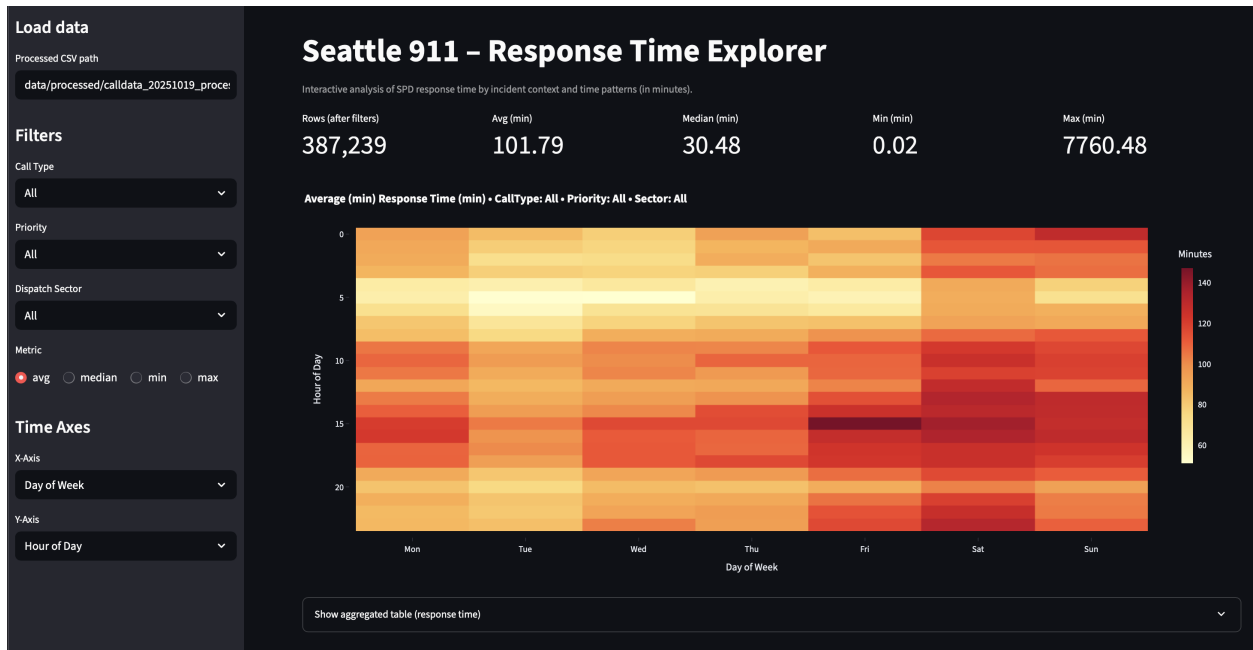


Figure 2: Interactive heatmap showing average, median, minimum, and maximum SPD response times (in minutes) across different temporal dimensions. The dashboard allows filtering by call type, priority, and dispatch sector to reveal spatial-temporal trends in emergency response efficiency.

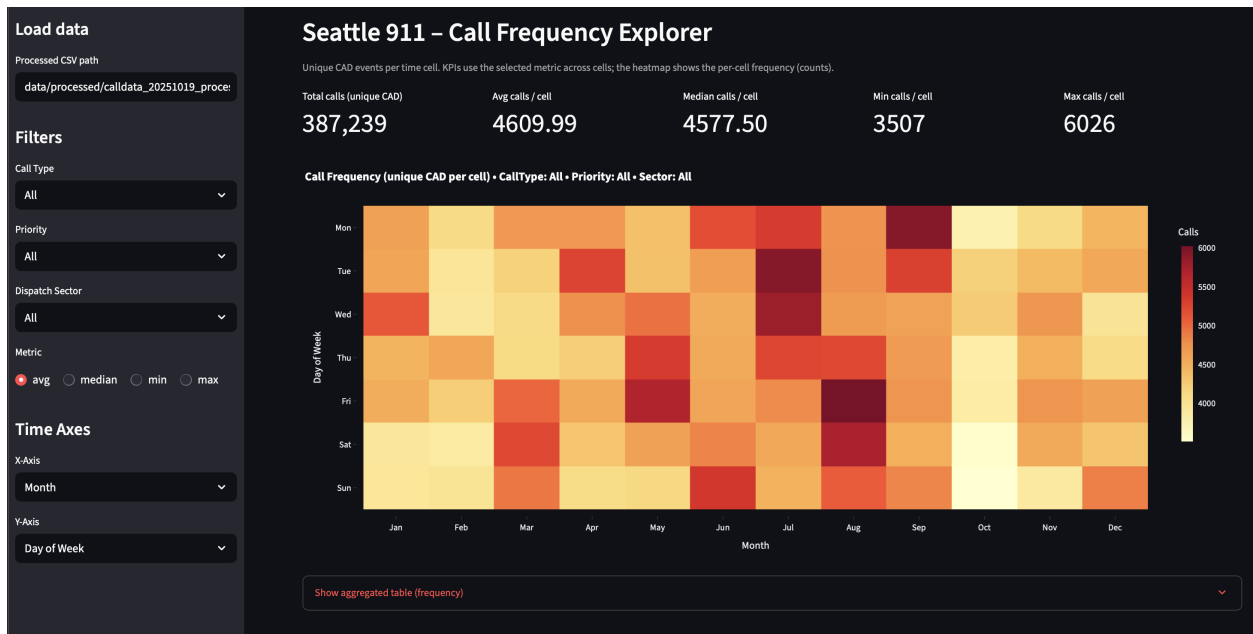


Figure 3: Interactive visualization illustrating the frequency of 911 calls (unique CAD events) by time of day, day of week, and month. The heatmap reveals cyclical patterns in call volume and demand concentration across temporal intervals.

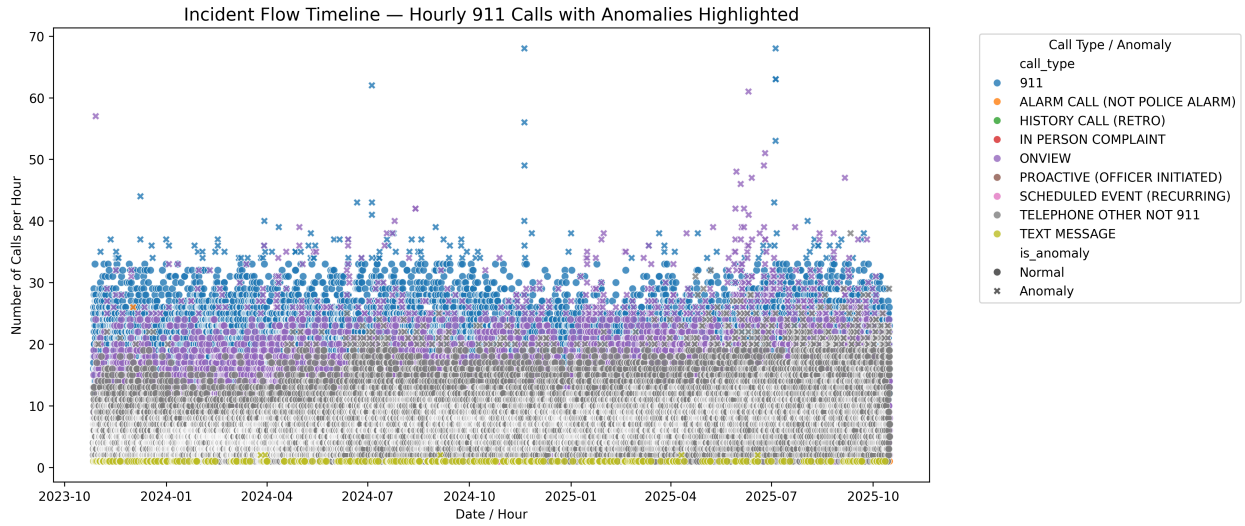


Figure 4: Scatterplot of hourly call volumes over time, with anomalies detected by the Isolation Forest model highlighted. This visualization identifies irregular spikes or drops in call activity that deviate from normal operational patterns, supporting system monitoring and anomaly analysis.